



PSLE Mathematics Practice

Primary 5 — Chapters 1 to 5

Chapter 1: Numbers to 10 Million

Chapter 2: Four Operations of Whole Numbers

Chapter 3: Fraction and Division

Chapter 4: Four Operations of Fractions

Chapter 5: Area of Triangle

Section A — 30 Multiple Choice Questions

Section B — 30 Open-Ended Questions

Answer Key on the last pages

Name: _____ Date: _____

Section A — Multiple Choice Questions

Choose the correct answer for each question. Questions 1–30.

Chapter 1: Numbers to 10 Million

1. What is the value of the digit 7 in 7,364,982?

- A) 7
- B) 70,000
- C) 700,000
- D) 7,000,000

2. Which of the following is the expanded form of 5,203,048?

- A) $5,000,000 + 200,000 + 3,000 + 40 + 8$
- B) $5,000,000 + 200,000 + 3,000 + 48$
- C) $5,000,000 + 203,000 + 48$
- D) $5,000,000 + 20,000 + 300 + 48$

3. Which number is the greatest?

- A) 4,982,310
- B) 4,928,310
- C) 4,983,100
- D) 4,829,999

4. Which of the following correctly arranges the numbers 6,145,200, 6,154,200 and 6,145,020 in ascending order?

- A) $6,145,020 < 6,145,200 < 6,154,200$
- B) $6,145,200 < 6,145,020 < 6,154,200$
- C) $6,154,200 < 6,145,200 < 6,145,020$
- D) $6,145,020 < 6,154,200 < 6,145,200$

5. Round 3,847,562 to the nearest hundred thousand.

- A) 3,800,000
- B) 3,850,000
- C) 3,900,000
- D) 4,000,000

6. Round 6,995,430 to the nearest ten thousand.

- A) 6,990,000
- B) 6,995,000
- C) 7,000,000
- D) 6,900,000

Chapter 2: Four Operations of Whole Numbers

7. $64 \times 3,000 = ?$

- A) 19,200
- B) 192,000
- C) 1,920
- D) 64,300

8. $72,000 \div 900 = ?$

- A) 8
- B) 80
- C) 800
- D) 0.8

9. $20 + 4 \times (7 - 2) = ?$

- A) 40
- B) 120
- C) 24
- D) 34

10. $(16 - 4) \div 3 + 2 \times 5 = ?$

- A) 14
- B) 10
- C) 26
- D) 8

11. A bakery bakes 18 trays of 24 muffins each. 45 muffins are sold. How many muffins are left?

- A) 387
- B) 432
- C) 342
- D) 477

12. A farmer has 250 chickens. He buys 4 times as many more chickens. How many chickens does he have in total?

- A) 1,000
- B) 1,250
- C) 1,500
- D) 750

Chapter 3: Fraction and Division

13. Express $16 \div 24$ as a fraction in its simplest form.

- A) $\frac{2}{3}$
- B) $\frac{4}{6}$
- C) $\frac{8}{12}$
- D) $\frac{1}{2}$

14. Which division is correctly expressed as the fraction $\frac{5}{9}$?

- A) $9 \div 5$
- B) $5 \div 9$
- C) $45 \div 9$
- D) $9 \div 45$

15. Express $20 \div 35$ as a fraction in its simplest form.

- A) $\frac{4}{7}$
- B) $\frac{5}{7}$
- C) $\frac{2}{7}$
- D) $\frac{4}{5}$

16. Express $\frac{5}{8}$ as a decimal.

- A) 0.625
- B) 0.58
- C) 1.6
- D) 0.85

17. Express $\frac{11}{4}$ as a decimal.

- A) 2.75
- B) 0.275
- C) 2.25
- D) 27.5

18. Which of the following fractions gives a terminating (exact) decimal?

- A) $\frac{1}{3}$
- B) $\frac{1}{6}$
- C) $\frac{3}{4}$
- D) $\frac{5}{6}$

Chapter 4: Four Operations of Fractions

19. $2\frac{3}{4} + 1\frac{5}{8} = ?$

- A) $4\frac{3}{8}$
- B) $3\frac{11}{8}$
- C) $3\frac{3}{8}$
- D) $4\frac{1}{8}$

20. $6\frac{1}{5} - 2\frac{3}{5} = ?$

- A) $3\frac{3}{5}$
- B) $4\frac{2}{5}$
- C) $3\frac{2}{5}$
- D) $4\frac{3}{5}$

21. $\frac{3}{5} \times \frac{2}{7} = ?$

- A) $\frac{6}{35}$
- B) $\frac{5}{12}$
- C) $\frac{6}{12}$
- D) $\frac{5}{35}$

22. $2\frac{1}{3} \times 6 = ?$

- A) 14
- B) 12
- C) $13\frac{1}{3}$
- D) 7

23. A container holds 72 marbles. $\frac{3}{8}$ of them are blue. How many blue marbles are there?

- A) 27
- B) 24
- C) 30
- D) 21

24. Ali spent $\frac{2}{5}$ of his money on a book and $\frac{1}{4}$ of the remainder on a pen. If he had \$60 at first, how much did he spend on the pen?

- A) \$9
- B) \$15
- C) \$24
- D) \$12

Chapter 5: Area of Triangle

25. In a triangle, the height is always:

- A) equal to one of the sides
- B) the perpendicular distance from the base to the opposite vertex
- C) the longest side of the triangle
- D) parallel to the base

26. A right-angled triangle has legs of 6cm and 9cm, and a hypotenuse of about 10.8cm. Which pair of lengths can be used directly as base and height?

- A) 6cm and 10.8cm
- B) 9cm and 10.8cm
- C) 6cm and 9cm
- D) any two sides

27. Find the area of a triangle with base 12cm and height 7cm.

- A) 42cm^2
- B) 84cm^2
- C) 19cm^2
- D) 35cm^2

28. A triangle has an area of 54cm^2 and a base of 12cm. Find its height.

- A) 9cm
- B) 4.5cm
- C) 6cm
- D) 13.5cm

29. A rectangle measuring 14cm by 8cm has a triangular piece (base 6cm, height 8cm) cut from one corner. Find the remaining area.

- A) 88cm^2
- B) 96cm^2
- C) 104cm^2
- D) 112cm^2

30. A figure is formed by a rectangle $10\text{cm} \times 6\text{cm}$ with a triangle (base 10cm , height 5cm) attached on top. Find the total area of the figure.

- A) 85cm^2
- B) 75cm^2
- C) 90cm^2
- D) 110cm^2

Section B — Open-Ended Questions

Show your working clearly. Questions 1–30.

Chapter 1: Numbers to 10 Million

1. Write 4,706,090 in expanded form.

2. State the value of the digit 8 in each of the following numbers: (a) 8,341,205 (b) 2,689,417 (c) 1,053,780

3. Arrange the following numbers in descending order: 9,340,215 / 9,304,521 / 9,340,152

4. The population of Town A is 2,845,600 and Town B is 2,854,600. Which town has the larger population? Explain using place values.

5. A company reports its annual revenue as \$7,462,850. Round this figure to the nearest million and to the nearest hundred thousand.

6. Explain why 3,995,000 rounds to 4,000,000 when rounded to the nearest hundred thousand, even though the digit in the hundred thousands place is 9.

Chapter 2: Four Operations of Whole Numbers

7. Find the value of $84 \times 5,000$.

8. Find the value of $63,000 \div 700$.

9. Find the value of $50 - 2 \times (8 + 3)$.

10. Find the value of $(24 \div 6 + 3) \times 4$.

11. A stationery shop orders 22 boxes of 36 pencils each. 150 pencils are found to be defective and returned. How many good pencils remain?

12. A cinema has 320 seats in Hall A. Hall B has 3 times as many seats as Hall A, minus 60 seats. How many seats does Hall B have, and how many seats are there altogether in both halls?

Chapter 3: Fraction and Division

13. Express $18 \div 30$ as a fraction in its simplest form.

14. Write the division sentence $27 \div 12$ as a fraction, then simplify it as far as possible (give your answer as a mixed number).

15. Express $7/16$ as a decimal.

16. Express $13/5$ as a decimal.

17. 21 sweets are shared equally among 28 children. What fraction of a sweet does each child get? Give your answer in its simplest form.

18. A ribbon of length 9m is cut into 8 equal pieces. Express the length of each piece (a) as a fraction of a metre in simplest form, and (b) as a decimal.

Chapter 4: Four Operations of Fractions

19. Find $4 \frac{1}{6} + 2 \frac{5}{6}$, giving your answer as a whole number or mixed number in simplest form.

20. Find $7\frac{2}{9} - 3\frac{5}{9}$.

21. Find $\frac{5}{6} \times \frac{9}{10}$, giving your answer in its simplest form.

22. Find $1\frac{3}{4} \times 8$.

23. A tank is $\frac{2}{3}$ full of water. After 20 litres are removed, it becomes $\frac{1}{3}$ full. Find the capacity of the tank.

24. Mrs Tan baked 84 cupcakes. She sold $\frac{3}{4}$ of them in the morning and $\frac{2}{3}$ of the remainder in the afternoon. How many cupcakes were left unsold?

Chapter 5: Area of Triangle

25. In triangle PQR, PQ is chosen as the base. Describe how you would find the height for this base, and state one property this height must have.

26. A triangle has a base of 18cm and an area of 108cm^2 . Find its height.

27. Find the area of a triangle with base 15cm and height 6.4cm.

28. The area of a triangle is 96cm^2 . If its height is 16cm, find its base.

29. A piece of land is shaped like a rectangle 20m by 12m, with a triangular corner (base 8m, height 12m) removed for a path. Find the area of the remaining land.

30. A composite figure consists of a rectangle 16cm by 9cm and a triangle with base 16cm and height 7cm attached to one side of the rectangle. Calculate the total area of the figure, showing your working.

Answer Key

Section A — Multiple Choice Questions

Chapter 1

1	D) 7,000,000 — the 7 is in the millions place.
2	A) $5,000,000 + 200,000 + 3,000 + 40 + 8$
3	C) 4,983,100 — comparing hundred-thousands (9) then ten-thousands (8) then thousands ($3 > 2$).
4	A) $6,145,020 < 6,145,200 < 6,154,200$
5	A) 3,800,000 — ten-thousands digit is 4 (< 5), round down.
6	C) 7,000,000 — thousands digit is 5, ten-thousands digit 9 rounds up and carries.

Chapter 2

7	B) 192,000 ($64 \times 3 = 192$, $\times 1000$)
8	B) 80 ($72,000 \div 900 = 720 \div 9 = 80$)
9	A) 40 ($7-2=5$; $4 \times 5=20$; $20+20=40$)
10	A) 14 ($12 \div 3=4$; $2 \times 5=10$; $4+10=14$)
11	A) 387 ($18 \times 24=432$; $432-45=387$)
12	B) 1,250 (bought $4 \times 250=1,000$; total $250+1,000=1,250$)

Chapter 3

13	A) $2/3$ ($16/24 \div 8$)
14	B) $5 \div 9$
15	A) $4/7$ ($20/35 \div 5$)
16	A) 0.625 ($5 \div 8$)
17	A) 2.75 ($11 \div 4$)
18	C) $3/4 = 0.75$ exactly; the others recur.

Chapter 4

19	A) $4 \frac{3}{8}$ ($2 \frac{6}{8} + 1 \frac{5}{8} = 3 \frac{11}{8} = 4 \frac{3}{8}$)
20	A) $3 \frac{3}{5}$ ($31/5 - 13/5 = 18/5$)

21	A) $6/35$
22	A) $14 (7/3 \times 6 = 42/3)$
23	A) $27 (3/8 \times 72)$
24	A) \$9 (book: $2/5 \times 60 = 24$; remainder 36; pen: $1/4 \times 36 = 9$)

Chapter 5

25	B) the perpendicular distance from the base to the opposite vertex
26	C) 6cm and 9cm — they meet at the right angle
27	A) $42\text{cm}^2 (1/2 \times 12 \times 7)$
28	A) 9cm ($54 = 1/2 \times 12 \times h$)
29	A) $88\text{cm}^2 (112 - 24)$
30	A) $85\text{cm}^2 (60 + 25)$

Section B — Open-Ended Questions

Chapter 1

1	$4,706,090 = 4,000,000 + 700,000 + 6,000 + 90$
2	(a) 8,000,000 (b) 80,000 (c) 80
3	$9,340,215 > 9,340,152 > 9,304,521$
4	Town B — millions and hundred-thousands digits are equal (2, 8), but Town B's ten-thousands digit (5) is greater than Town A's (4).
5	Nearest million: \$7,000,000. Nearest hundred thousand: \$7,500,000.
6	The ten-thousands digit is 9 (≥ 5), so it rounds up. This causes the hundred-thousands digit to carry over into the millions place, giving 4,000,000.

Chapter 2

7	420,000
8	90
9	$50 - 22 = 28$
10	$(4+3) \times 4 = 28$
11	$22 \times 36 = 792$; $792 - 150 = 642$ good pencils
12	Hall B: $3 \times 320 - 60 = 900$ seats. Total: $320 + 900 = 1,220$ seats.

Chapter 3

13	$18/30 = 3/5$
14	$27/12 = 9/4 = 2\frac{1}{4}$
15	$7 \div 16 = 0.4375$
16	$13 \div 5 = 2.6$
17	$21/28 = 3/4$
18	(a) $9/8 = 1\frac{1}{8}$ m (b) $9 \div 8 = 1.125$ m

Chapter 4

19	$4\frac{1}{6} + 2\frac{5}{6} = 6\frac{6}{6} = 7$
20	$65/9 - 32/9 = 33/9 = 3\frac{2}{3}$
21	$45/60 = 3/4$
22	$7/4 \times 8 = 56/4 = 14$
23	$2/3 - 1/3 = 1/3$ of tank = 20 l, so full capacity = 60 litres
24	Morning: $3/4 \times 84 = 63$; remainder = 21; afternoon: $2/3 \times 21 = 14$; left unsold = $21 - 14 = 7$ cupcakes

Chapter 5

25	Draw a line from vertex R perpendicular to line PQ (extending PQ if necessary). This height must form a right angle (90°) with the base PQ.
26	$108 = \frac{1}{2} \times 18 \times h \rightarrow h = 12\text{cm}$
27	$\frac{1}{2} \times 15 \times 6.4 = 48\text{cm}^2$
28	$96 = \frac{1}{2} \times b \times 16 \rightarrow b = 12\text{cm}$
29	Rectangle: $20 \times 12 = 240$; triangle: $\frac{1}{2} \times 8 \times 12 = 48$; remaining = $240 - 48 = 192\text{ m}^2$
30	Rectangle: $16 \times 9 = 144$; triangle: $\frac{1}{2} \times 16 \times 7 = 56$; total = $144 + 56 = 200\text{cm}^2$